

Tinnitus (adults and children)

1. Assessment and investigations

NICE NG 155, 2020, NEJM 2018;378:1224



Red Whale

GEMS

Guidelines & Evidence Made Simple

Tinnitus is common, but for most there is no underlying significant pathology.

- Risk increases with: age, noise exposure, drugs: aminoglycosides, NSAIDs, diuretics.
- It can have a profound effect on wellbeing and mental health. Be mindful of suicide risk – ask about this.
- Education ('it can be horrible but is not harmful') and support to live with it are a critical part of care.

Tinnitus is usually considered to be a SUBJECTIVE experience, but NICE also describes OBJECTIVE tinnitus: this is tinnitus which can be heard by clinician, e.g. turbulent flow in carotid artery/jugular vein or rhythmical clicking of myoclonus in palatal/middle ear muscles. I suspect many of us wouldn't actually call that tinnitus, but it is something to be aware of.

Assessment

- **Clinical assessment for features that might warrant referral and RED FLAGS** (see referral criteria overleaf).
- NICE does not cover what to examine but a NEJM review suggests:
 - Assess hearing with Rinne and Weber's test and the whisper test.
 - Whisper test: stand arm's length behind patient. Occlude one ear. Whisper 3 letters/numbers and ask patient to repeat them back. 2 attempts are allowed (use different letters). Repeat in both ears.
 - Check cranial nerves for focal neurological signs.
 - Examine ears for reversible causes like wax or infection!
 - **If vertigo is present:** consider Meniere's, acoustic neuroma or migraine (see separate articles).
- **Assess the impact of tinnitus** on quality of life, sleep and mental health.
- **Look for and manage any anxiety or depression.**

NICE suggested that questionnaires can be used to assess tinnitus. It isn't clear whether it meant in primary or secondary care, but they are complex and we therefore suspect they will only be used in secondary care:

- Tinnitus Functional Index. Complicated to score, covers a variety of domains.
- Tinnitus Questionnaire (52 items with the shorter mini-TQ having a mere 33 items to complete!).

Investigations (will be done in secondary care (if needed) if referred)

- **Audiological assessment** (assessment of hearing loss and communication needs). Hearing loss should be managed as per NICE hearing loss guidance (summarised in a separate GEMS).
- Hearing aids are not indicated for tinnitus with NO hearing loss (see management details overleaf).
- **Consider tympanometry if:**
 - Middle ear/eustachian tube dysfunction suspected.
 - Conductive hearing loss may be contributing to tinnitus.
- Do not offer: acoustic reflex testing, uncomfortable loudness testing or otoacoustic emissions tests (unless needed because of other signs or symptoms), or psychoacoustic tests (pitch and loudness matching).
- **Imaging: the scan of choice is usually an MRI of the internal auditory meatus** (we are assuming this is a secondary care decision, and that scanning is only indicated in those who meet the referral criteria).
 - **Those with symmetrical non-pulsatile tinnitus do NOT need a scan:** low chance of significant pathology.

Pulsatile tinnitus	Synchronous pulsatile tinnitus (tinnitus synchronised with pulse). May do MR angiogram/wider MRI or enhanced CT scan if glomus tumour possible (highly vascular tumours that can occur in middle ear/jugular vein/carotid artery).	Consider scan
	Non-synchronous pulsatile tinnitus.	Consider MRI
Non-pulsatile tinnitus	Non-pulsatile tinnitus WITH neurological/otological/head and neck signs.	Offer MRI
	Unilateral/asymmetrical non-pulsatile tinnitus WITHOUT neurological/otological or head and neck signs:	Consider MRI
	Symmetrical non-pulsatile tinnitus: low chance of significant pathology.	Do NOT scan

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2. Management and referral

NICE NG 155, 2020

Management

• Information for patients

Tinnitus is common and may resolve by itself (40% of mild tinnitus and 20% of severe tinnitus resolved by 5y).

- It commonly occurs with hearing loss but is rarely associated with other underlying health problems.
- Symptoms can wax and wane over time, and can be made worse by stress and loud noise.
- There are strategies to help people live well with tinnitus.
- Local and national support groups are available: British Tinnitus Association: <https://www.tinnitus.org.uk/>

• **All patients should be offered 'tinnitus support'**. NICE describes this as a "2-way process of information-giving and discussion to help the healthcare professional understand the difficulties and goals of the person. A management plan is jointly developed and the person is supported to follow the plan".

• **Refer if they meet any of the criteria set out below.**

• Hearing aids:

- Offer a hearing aid if hearing loss affects ability to communicate.
- Consider a hearing aid if hearing loss but no communication difficulties.
- Do NOT offer a hearing aid if tinnitus but NO hearing loss.

• **Psychological therapies.** If 'tinnitus support' insufficient, offer, in this order:

- Digital tinnitus-focused CBT.
- Group-based interventions (often include mindfulness, CBT, acceptance commitment therapy).
- Individual tinnitus-focused CBT.

• NICE was unable to recommend sound therapy (white noise/music) or neuromodulation (usually using in-ear-sound generating device). Lack of evidence. Research of effectiveness needed.

• Do NOT use betahistine.

Referral criteria and timeframes

Immediate referral if:	Tinnitus with significant suicidal risk.	
	Tinnitus with sudden-onset significant neurological symptoms/signs.	
	Tinnitus with acute uncontrolled vestibular symptoms (e.g. vertigo).	
	Tinnitus with sudden-onset hearing loss.	
Tinnitus with neurological symptoms		
Tinnitus with sudden-onset significant neurological features/suspected stroke		Immediate referral
Tinnitus with acute uncontrolled vestibular symptoms (e.g. vertigo)		
Tinnitus with hearing loss		
Tinnitus with recent sudden-onset hearing loss (developed over $\leq 3d$ in past 30d)		Refer: to be seen within 24h
Tinnitus with less recent sudden-onset hearing loss (sudden onset but $>30d$ ago)		Refer: to be seen within 2w
Tinnitus with rapidly worsening hearing loss over past 4–90d		
Tinnitus with unilateral or asymmetrical hearing loss		Routine referral
Tinnitus affecting mental wellbeing		
Tinnitus with significant suicidal risk		Immediate referral
Tinnitus with mental distress preventing them carrying out their usual daily activities (if simple first-line interventions insufficient)		Refer: to be seen within 2w
Tinnitus bothering people if first-line interventions have not been sufficient		Routine referral
Persistent tinnitus		
Persistent OBJECTIVE tinnitus (<i>objective = clinician can hear, e.g. turbulent flow in carotid artery/jugular vein or rhythmical clicking of myoclonus in palatal/middle ear muscles</i>)		Routine referral
Persistent pulsatile tinnitus		Possible routine referral
Persistent unilateral tinnitus		